HWK 6

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> Q6. Write your own function starting from the code above that analyzes protein drug interactions by reading in any protein PDB data and outputs a plot for the specified protein. (See class lecture 9 for further details).

Documentation

The input is my__protein this is a string and it contains a PDB ID

Role of the function: it reads a PDB file, extracts the chain A alpha-carbon atoms, from there it extracts specifically the B factor values to which it then plots the said B-factors for the specified protein which is what plotb3() does. It returns the B-factor as a numeric vector while displaying the B-factor plot. In order to use it, one must call the function with the PDB ID

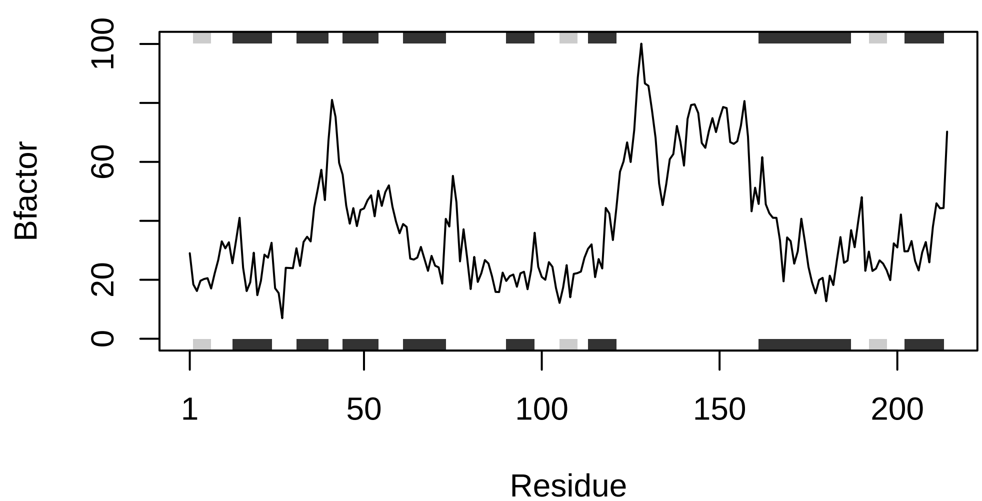
Output: it returns a numeric vector of B-factor values and produces the b-factor plot for a specific protein.

```
library(bio3d) #load R package which is bio3d
```

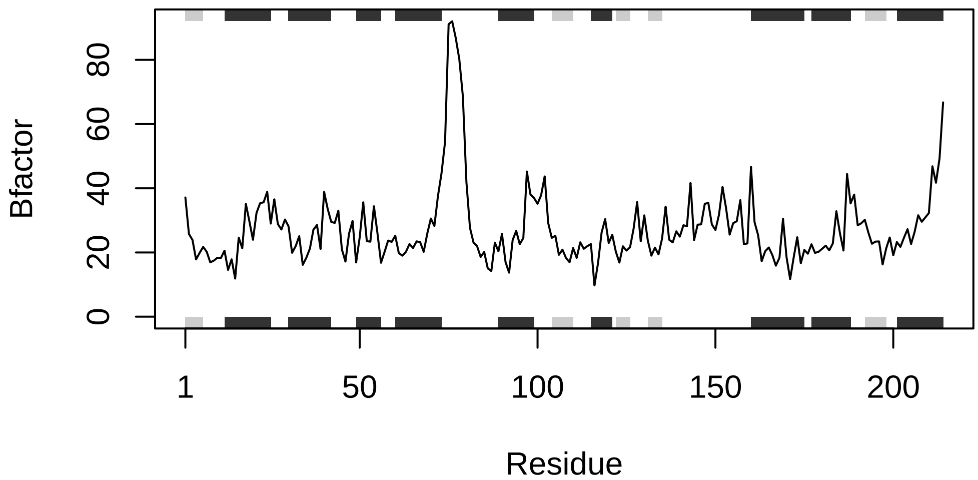
```
#plot_bfactor is the name of the function that takes one input called my_protei

plot_bfactor <- function(my_protein) {

#Read.pdb() reads the PDB structure file
  pdb <- read.pdb(my_protein)

#Extract chain A alpha carbons using trim.pdb() to filter a pdb structure
#Elety gives us the element for one atom per residue.
  chainA <- trim.pdb(pdb, chain = "A", elety = "CA")

#Extract B-factors, chainA is the trimmed protein structure
#$atom is the operator to extract data frame inside chain A
#$b accesses the b column and thus B-factor value
  b <- chainA$atom$b
```

```
#Plot B-factors
#b- first argument and numeric vector of B-factor values
#sse=chainA draws structural elements
#typ = "l" creates a line plot
#ylab= "Bfactor" sets the y axis
  plotb3(b, sse = chainA, typ = "l", ylab = "Bfactor")

#Return B-factors
  return(b)
}

#These are the PDB ID strings within my_protein
my_proteins <- c("4AKE", "1AKE", "1E4Y")
for (protein in my_proteins) {
  plot_bfactor(protein)
}
```

```
## Note: Accessing on-line PDB file
```




```
## Note: Accessing on-line PDB file
##  PDB has ALT records, taking A only, rm.alt=TRUE
```



Note: Accessing on-line PDB file